NPRB2301 Sample and Data Update

January 7, 2026

This report summarizes the samples available for NPRB 2301, their status and the status of laboratory processes. Updates will be made available whenever significant updates to the sample database or analytical data become available.

Version Notes

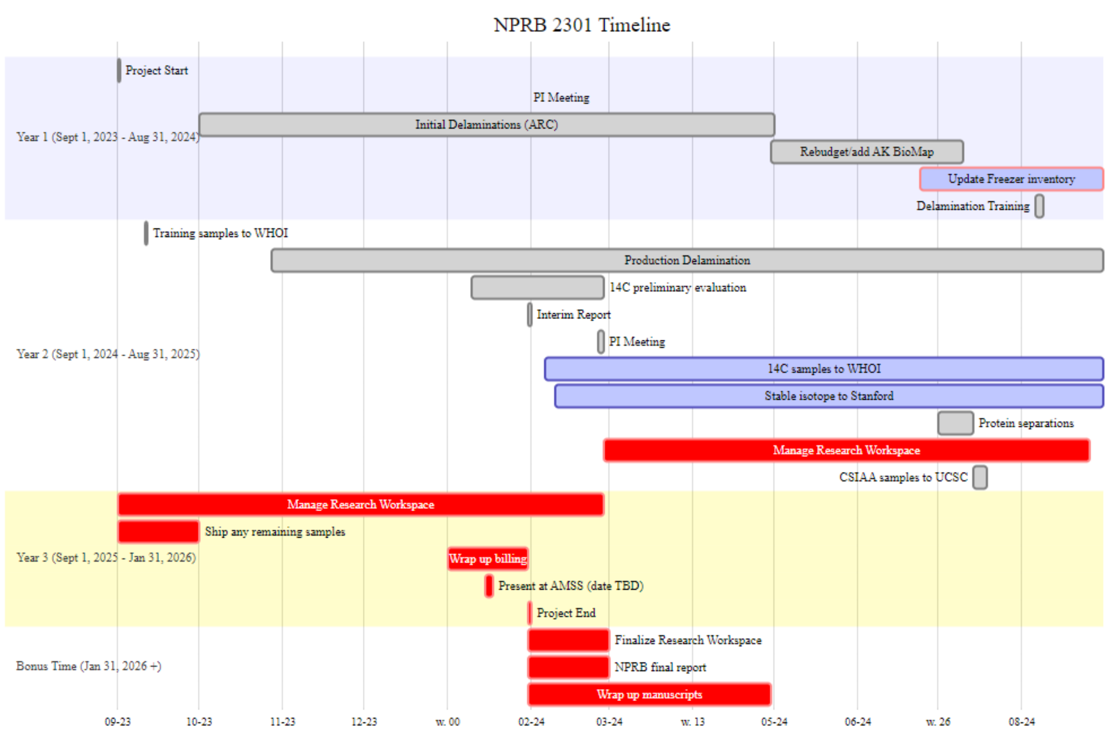
- New columns added to make data management easier:
  - "new_layer_type" is a more accurate reflection of what the layers contained and is standardized to minimize categories and confusion
  - "std_lyr_id" takes all the different formats of layer_id and ams_id and makes a single format across ALL samples.
  - "sample_desc" Text descriptor of what type of sample it is and from what species to make filtering quicker in analytics
  - "protein_type" is for filtering whole layers, soluble or insoluble portions

- All soluble/insoluble protein samples have been returned. Data can be found in NPRB2301_layer_results, 14C_layer_results sheet. Filter on protein type.
- Lens diameters have been updated and double checked. Charlotte and Cindy are working on taking more measurements from eyes still in the freezer. Data can be found in NPRB2301_layer_results, layer_diameter_results sheet.
- Beth, Charlotte and Cindy are working on the length extrapolation, stay tuned for a mini-report.
- All spiny dogfish ages have been finalized

Links and Resources

- github repo (public)
- Research Workspace (password protected)
- Google Folder (access controlled)

Project Timeline



Specimen, Sample and Layer Summaries

- Each animal has a specimen_ID. The exception is spiny dogfish embryos, which are considered samples taken of the adult.
- Each eye, embryo, spine, etc. has a unique sample_id, which links back to the specimen_id that that sample was taken from.
- Each eye is further separated into layers and each layer_id links back to the sample_id (and thus, specimen_id).Layers are numbered sequentially from the outside towards the center. Early numbering convention used "N" for nucleus, and "R" for the remainder.
- Embryo each have a sample_id, but two eyes. Each eye and/or layer is labeled with the sample_id, L or R and layer number (as applicable).

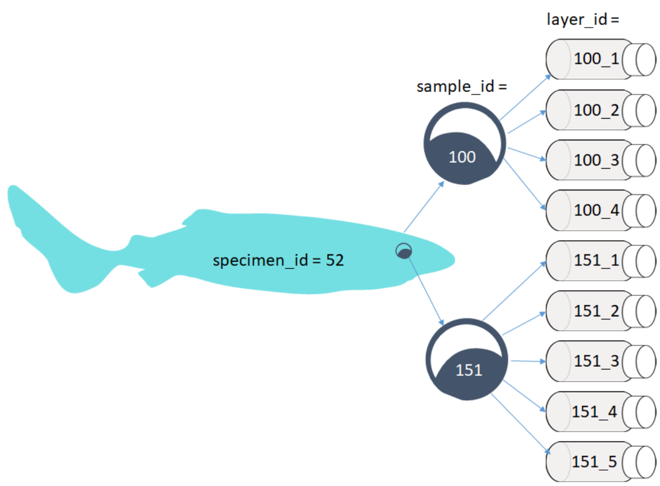


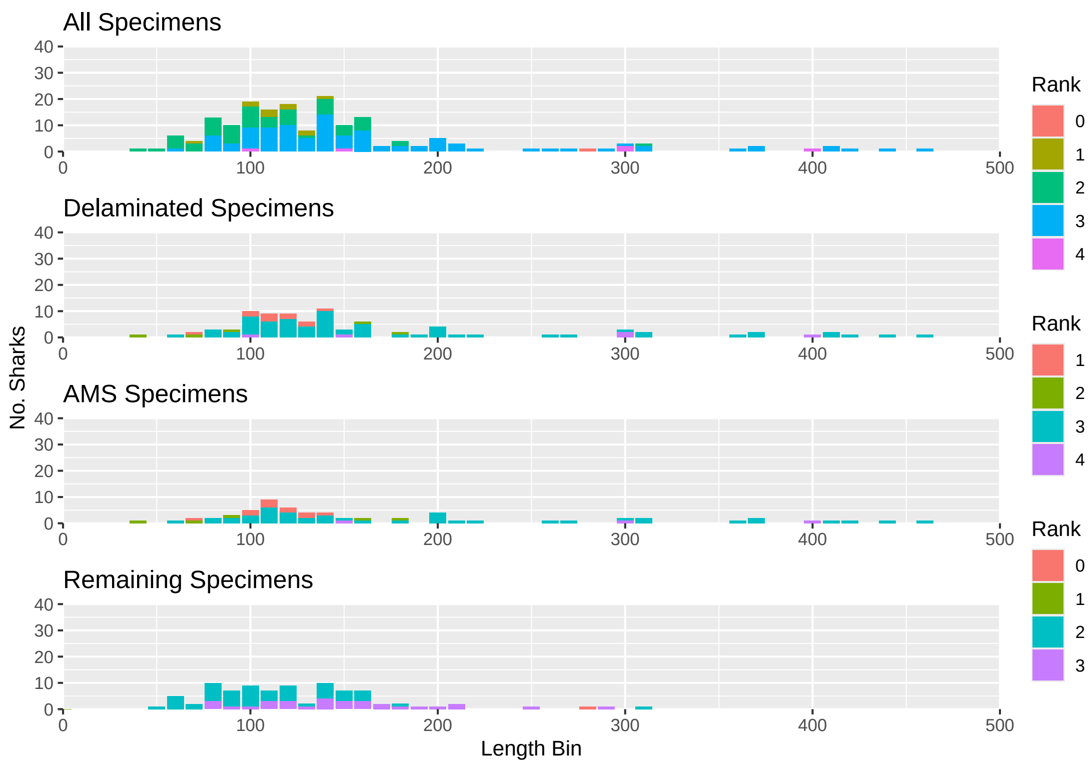
Figure 1: Specimens to samples to layers flow chart

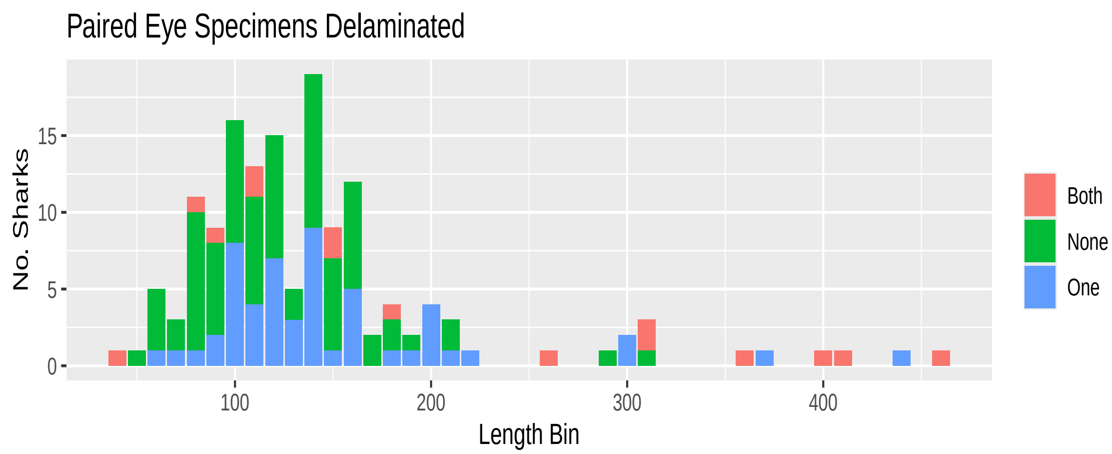
Pacific sleeper sharks

The below table only includes specimens with both length and haul data. There are some samples without haul data, and some without length. As data gaps get filled in, these numbers will change. Please request any other data breakdowns for future reports.

N PSS	N PSS eyes	N PSS Delaminated	N PSS Layers
174	322	115	701

For the purposes of the below graphs, all unknown length types are treated as Total Length until we resolve how to determine that. Ranks are from 1 (lowest) to 4 (highest) and described here. Rank = 1 are samples that had poor handling and may have thawed before delamination, Rank = 2 are collections with missing length or haul data, but do have at least a "source" (being who provided the sample), Rank = 3 are collections with complete data, but some uncertainties (i.e., length type), and Rank = 4 are survey collections with best chain of custody and full data available. Many specimens have not yet been attributed to a "source" in the database, this is intentional. Many of the at-sea observer provided samples did not have full data with the sample, or deck forms submitted. Some of these data gaps will be filled as we dig through at-sea observer databases, and those specimens will change from rank = 0 to the appropriate rank 1 - rank 4 value.





Spiny dogfish

N SD	N SD with full data	N SD available	N SD with samples but no length
28	17	12	5

The above numbers include samples sent from Washington. There are some BC samples that have not been processed.

Carbon 14

The estimated N 14C samples left per budget reflects those that have been submitted, but do not reflect the lost samples or the potential rebudgeting of WHOI funds. We are going to have some remaining samples available and need to discuss how best to proceed and how fast we can proceed.

N 14C sent	N 14C returned	N 14C samples left per budget	Total N budgeted
515	523	45	549

The below table summarizes the number of each species which have 14C results from at least one layer

N 14C PSS	N 14C SD
64	9

14C results can be found in NPRB2301_layer_results.

Date Sent	N layers sent	N layers returned
Mar 25, 2019	11	11
Aug 5, 2019	8	8
Sept 11, 2024	36	35
Feb 6, 2025	70	70
Mar 31, 2025	96	95
Apr 28, 2025	50	50
July 14, 2025	68	68
Aug 25, 2025	36	30
Sept 9, 2025	141	141
Dec 2, 2025	18	14
Dec 15, 2025	68	0

Stable Isotopes

N SIA sent	N SIA returned	N SIA samples left per budget	Total N Budgeted
506	506	54	560

Date Sent	N layers sent	N layers returned
Feb 4, 2025	70	70
Mar 31, 2025	107	107
Apr 1, 2025	15	15
Apr 25, 2025	35	35
July 14, 2025	68	68
Sept 19, 2025	211	209

There are two samples not yet reported in the data yet, pending more information. Budget numbers are being confirmed for any potential remaining funds.

Stable isotope results can be found in NPRB2301_layer_results.

CSIA-AA

NOTE: we have funds for more samples than initially planned. Total sample size is now 35. The samples were so "clean" that they did not cost as much to process. We need to discuss how best to proceed with the remaining 5 samples: should we do more samples from an individual already sent, or a new shark? These remaining 5 samples will be pre-billed so we have some flexibility on sending these.

N CSIA_AA sent	N CSIA_AA returned	N CSIA_AA samples left per budget	Total N Budgeted
29	25	30	30

Protein Separation

All of the 14C data for soluble and insoluble proteins are back from NOSAMS. 54 samples were submitted, with data returned for 45. Eight of the samples without data were soluble portions that were too small and the other was lost during processing.

Soluble and insoluble portion layer results are in the NPRB2301_layer_results 14C_layer_results sheet. Samples for this analysis are noted by std_layer_ids with "I" or "S" at the end and can be easily filtered by protein_type.

Spiny Dogfish Ageing

WDFW has concluded the final reading of spiny dogfish spines. For reference, the spine reading methods are detailed in Tribuzio et al. 2017.

Spiny dogfish ages are available in NPRB2301_dogfish_spine_ages_PRELIMINARY. The age estimates follow the estimation methods recommended by Tribuzio et al. 2010.